Muungano: Decentralized Zero-Knowledge Information Signaling over Interledger and Open Payments

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Abstract

Millions of individuals and micro-, small-, and medium-sized enterprises (MSMEs) across Africa remain underserved by formal lenders despite their active, high-volume participation in the digital economy. While mobile money transactions, digital wallet logs, and merchant payment activities offer deep behavioral insights into real-time cash flows and financial stability, this valuable data remains fragmented across siloed ledger systems, prone to selective disclosure fraud, or locked behind centralized data honeypots.

In this paper, we introduce **Muungano**, an open, decentralized financial trust infrastructure built natively on the Interledger Protocol (ILP) and the Open Payments architecture. Muungano transforms isolated, multi-provider transactional events into a trusted, highly portable, and sovereign financial reputation layer. By leveraging the Grant Negotiation and Authorization Protocol (GNAP) to securely bridge and verify authoritative ledger domains via Web-standard Wallet Addresses, the protocol shifts the verification burden entirely client-side.

We implement a lightweight, parameterizable Poseidon-based Sparse Merkle Tree (SMT) circuit utilizing PLONKish arithmetization over the BN254 scalar field. When a lender issues a financial risk query tied to an Open Payments Quote, the consumer's local device acts as a prover—mathematically demonstrating that their aggregate financial standing meets or exceeds the necessary credit risk thresholds without revealing their underlying transaction history or private identifiers. Because the resulting zero-knowledge range proofs are highly compressed, the entire cryptographic payload easily fits within standard network Maximum Transmission Unit (MTU) boundaries, guaranteeing sub-millisecond, out-of-band verification at the ILP routing layer.

By replacing expensive, manual document auditing and centralized registries with an instantaneous, privacy-preserving cryptographic conversation, Muungano provides a robust, scalable, and secure engineering foundation for the next generation of credit-extended open banking across emerging payment ecosystems.

Keywords: PLONKish Arithmetization, Interledger, Open Payments, Zero-Knowledge Proofs, Mobile Money, GNAP, Financial Inclusion.

1 The ILP Transport Bottleneck: Nature of STREAM vs. ZK Proofs

The Interledger Protocol (ILP) stack leverages the **STREAM** transport protocol to execute high-throughput packetized micro-payments. STREAM operates by dividing a transactional request into a continuous flow of small, independent packet frames. Each STREAM packet is structurally bounded by strict network Maximum Transmission Unit (MTU) constraints—typically limited to 1500 bytes to prevent packet fragmentation, transmission retries, and latency spikes across multi-hop routing paths.

Conversely, Zero-Knowledge succinct proofs (such as SNARK proofs) generated over cryptographic curves represent significant serialization overhead. Even highly optimized proofs can occupy several kilobytes. Attempting to inject ZK proofs directly into the standard STREAM packet headers creates a severe performance impedance mismatch:

1. **Fragmentation Overhead:** A proof payload exceeding the network MTU limits forces the IP layer to fragment packets, causing intermediate routing connectors to experience high memory consumption and latency.
2. **High Settlement Latency:** Streaming payment connectors require sub-millisecond execution loops. Verifying mathematical proof constraints on every micro-packet routing decision renders stream settlement slow and cost-prohibitive.

To resolve this mismatch, Muungano departs from in-stream packet verification. Instead, we introduce a separation of concerns: routing layers execute lightweight financial transfers, while cryptographic verification is shifted **out-of-band** to the API authorization layer.

2 Pre-Payment Signaling: The GNAP Authorization Handshake

Rather than verification during active packet streaming, Muungano executes the ZK handshake out-of-band during the payment authorization phase using the **Grant Negotiation and Authorization Protocol (GNAP)**. GNAP is highly flexible, allowing client wallets to negotiate access tokens by presenting customized cryptographic assertions.

The signaling protocol utilizes the standard Open Payments challenge structure. When a borrower requests a credit line quote, the lender's wallet generates an Open Payments **Quote Resource** and returns it with a custom challenge header:

```
X-Muungano-Challenge: <Quote_ID>
```

Additionally, the lender generates a Hash Time-Lock Contract (HTLC) lock condition:

$$H = \mathrm{SHA256}(S)$$

where S is a private, random 32-byte preimage (the secret promise) generated by the lender. The lock H binds the payment to this specific quote request.

The client wallet intercepts this challenge, aggregates credit scores from connected ledgers (e.g., M-Pesa and local bank balances), and generates a ZK proof. Instead of attaching this proof to subsequent STREAM packets, the client submits the proof to the lender's GNAP endpoint. Once verified, the lender authorizes the transaction and releases the secret preimage S, allowing the payment to settle atomically.

3 Mathematical Foundations & PLONKish Arithmetization

Muungano represents ledger relationships and constraints within a PLONKish arithmetization grid over the BN254 scalar field $\mathbb{F}_r$, where:

$r = 21888242871839275222246405745257275088548364400416034343698204186575808495617$

3.1 PLONKish Execution Matrix

The constraint layout utilizes advice columns $\mathbf{A}$, fixed columns $\mathbf{F}$, and selector columns $\mathbf{q}$. Range checks and inequality constraints are optimized using UltraPLONK-style lookup tables. To prove that the borrower's credit score meets the threshold without revealing the scores, we enforce the gate relation:

$$q_{\text{range}} \cdot (S_{\text{total}} - S_1 - S_2) = 0$$

where S_1 and S_2 are private score witnesses from independent ledgers. We then verify:

$$T < S_{\text{total}} + 1$$

using a lookup table to assert that the difference fits within 34 bits, preventing overflow attacks.

3.2 Poseidon Merkle Membership Structure

To verify the authenticity of the scores S_1 and S_2 without exposing identity, we compute membership paths within two independent parameterizable d-depth SMT registries (configured as $d = 4$ for our experimental prototype; e.g., Tree 1 for Mobile Money, Tree 2 for Bank). The circuit decomposes the indices into direction bits b_i:

$$b_i \in \{0, 1\} \implies \text{direction\_bits} = \text{num\_to\_bits}(\text{index}, d)$$

and climbs to the respective public roots R_1 and R_2 utilizing the Poseidon duplex sponge. Crucially, both tree climbs share the same identity witness W_{identity}, forcing cross-ledger identity binding.

The complete parallel validation flow, active sibling climbers, and the corresponding copy constraints are illustrated in Figure 1.

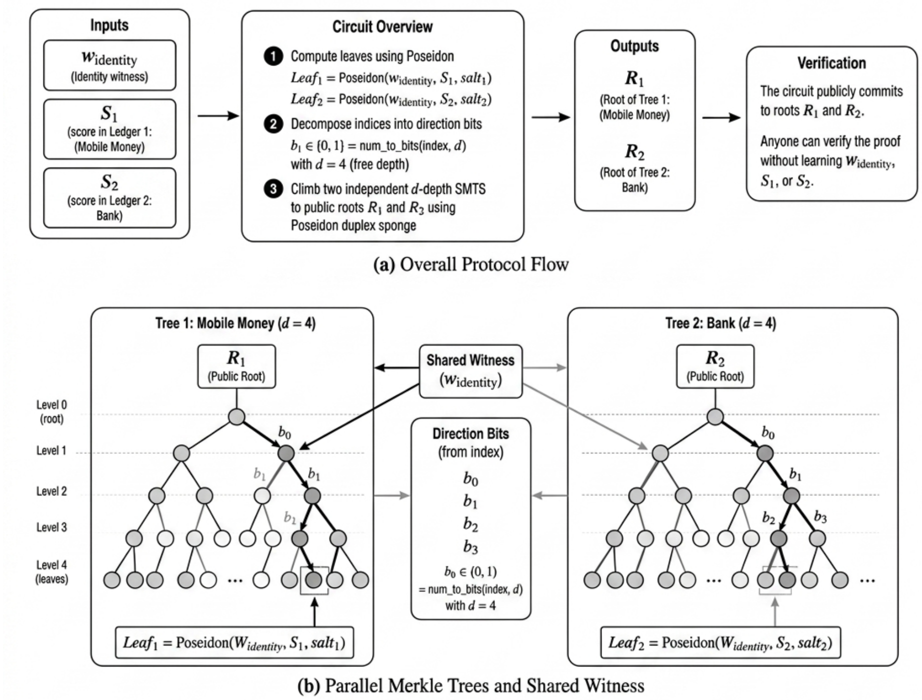

Table [Y]: Multi-Ledger Copy Constraints

Advice A	Advice B	Selector q	Enforced Constraint
$W_{identity}$	$W_{identity}$	q_{tree1}	Poseidon($W_{identity}$, S_1, $salt_1$) = $Leaf_1$
$W_{identity}$	$W_{identity}$	q_{tree2}	Poseidon($W_{identity}$, S_2, $salt_2$) = $Leaf_2$

(c) Table: Copy Constraints

Figure 1: Muungano Multi-Ledger SMT Circuit Architecture and Constraints.

4 The Muungano Protocol Execution Flow

The complete end-to-end cryptographic signaling loop is structured as follows:

1. **Quote & Lock Request:** The client requests a credit quote. The lender generates a secret preimage S, computes the lock $H = \text{SHA256}(S)$, and returns the quote challenge containing H.

2. **ZKIP Local Proving:** The client app aggregates private scores S_1, S_2, builds the proof π bound to the public inputs (R_1, R_2, T, H), and returns it via GNAP.

3. **Out-of-Band Verification:** The gateway verifies π. If valid, it forwards the payment packet locked with H.

4. **Atomic Settlement:** The lender reveals the secret preimage S to unlock and settle the transaction.

5 Conclusion & Market Incentives

Muungano introduces a paradigm shift in financial verification by moving the computing burden entirely client-side. Under this architecture, the zero-knowledge proof (π) is compiled locally on the borrower's mobile device (phone) using sandboxed ledger API connections. Consequently, raw financial data—such as individual transaction histories, bank balances, credit reference records, or private identifiers—never leaves the local storage boundary of the user's device. Neither the lender nor the Muungano signaling gateway ever interacts with, transfers, or stores any raw customer records.

This localized proving process fundamentally eliminates the "data honeypot" problem. Traditional credit reference registries represent highly lucrative targets for exploitation; by decentralizing proof generation to the edge, the system reduces the target vector to zero. Furthermore, this design aligns the commercial incentives of traditional banks, telecom providers, and credit registries: instead of acting as centralized database curators, they act as cryptographic state publishers. Financial institutions monetize their roles by selling signed Merkle state roots (needed to compile proofs) directly to consumers. This model preserves consumer privacy, eliminates data liability, and unlocks new query-based revenue channels for ledger providers, building a resilient foundation for global, privacy-preserving credit routing.

References

[1] A. Gabizon, Z. Williamson, and O. Ciobotaru. PLONK: Permutations over Lagrange-bases for Oecumenical Non-interactive arguments of Knowledge. *IACR Cryptology ePrint Archive*, Report 2019/953, 2019.

[2] S. Thomas and E. Schwartz. A Protocol for Interledger Payment Routing. *Interledger Architecture Board (IL-RFC 1)*, 2017.

[3] J. Richer and Y. Atkins. Grant Negotiation and Authorization Protocol (GNAP). *IETF RFC 9635*, 2024.